

technologies to OLED-based technology. An OLED-based display generally helps the designers in terms of having a better contrast, it is lightweight and it consumes less power. If you see the viewing angle as well, it is better in OLED as compared to TFT."

Flexible and curved OLED modules. Special OLED solutions are available for design engineers, making it possible to have curved and flexible displays for the futuristic designs. It is one of the major and important innovations in the display world, helping to design better and efficient displays. The popular Samsung YOUM line of displays targets designers of smart-phones and wearables.

Curved surface significantly reduces, and sometimes even eliminates, reflections from ambient light sources. This not only improves screen readability and image quality but also allows the displays to run at lower brightness, thereby increasing their power efficiency and battery-running time. These OLEDs, being incredibly thin at just a fraction of a millimetre, along with the curved and flexible surfaces, are a suitable choice for mobile displays and the trending wearable displays.

Even higher resolution and brightness. As if resolution on current Samsung Galaxy S5 wasn't enough, many vendors have introduced even-higher-resolution AMOLED panels and modules. One example is AUO's WQHD AMOLED panel with 2560×1440 resolution at 513PPI. But vendors like Sharp, Samsung and SDC have gone further and announced resolutions of 560, 664 and even 860PPI. Apart from high resolution, ultra-slim designs up to 0.57mm have been realised and special drivers are being integrated to increase their touch sensitivity.

Using new structure and new OLEDs, manufacturers have come up with high-brightness PMOLED modules, which mainly target the increasing demand for wearable devices. These new OLED modules offer unparalleled clarity, even in direct sunlight, featuring a luminance of 1000 nits, which is nearly ten times of that

Considerations while selecting a display

The difference between a successful project and a failed one lies in choosing the right screen for the project's environment. Selecting a display with right combination of technologies is the key to a successful design. You might want to consider the following technologies while selecting your display:

1. **Super-high brightness.** Demand for indoor and outdoor HMLs is growing and outdoor environments require much greater level of luminance in TFT-LCD modules than the conventional level of 300-500 nits for indoor environments. Super-high-bright TFT-LCD modules in the luminance range of 1000-1500 nits might be the right solutions for your outdoor products.
2. **Optical bonding.** Industries such as medical, transportation, avionics, factory automation, point of sale, automotive and marine require the display modules to have some kind of protection. Screens have protective-layer optically bonded to them, improving impact strength, thermal performance and visibility by reducing reflections.
3. **Anti-reflection front surface.** In outdoor environments, elimination of surface reflections is very important. Anti-reflection front surface in screen helps to improve visibility and contrast outdoors in direct sunlight.
4. **Transflective modules.** These modules have both transmissive and reflective modes working together for advanced performance. If you are designing a product to be operated in both brighter and darker environments, using a transflective module can be of great help.
5. **Operating temperature.** Display modules are used in a wide variety of applications where severe cold or heat may have to be faced. Modules are available with a wide range of operating temperatures, and selecting the one with appropriate range is the key.

available with average OLED modules.

The only challenge for OLED technology is the manufacturing cost and yield. Though OLEDs have many benefits associated with them, manufacturing of OLEDs is still lacking behind in terms of productivity. Nate Srinath, founder-director, Inxee says, "Cost of manufacturing OLEDs is much higher than LCDs, and also people still don't appreciate their significance. The major challenges for OLEDs are reducing production costs and increasing production capacities."

Light capturing cavities in numeric displays

Demand for compact and efficient products has resulted in more compact and space-saving designs in 7-segment numeric displays, like those found in Lumex LDP-2R2608RD-50 series. These latest LED-based offerings use the brightest chip technology and light-capturing cavities, which allows them to offer maximum brightness and legibility without sacrificing any colour intensity or uniformity.

These single-package designs with multiple digits and icons can display any type of word or icon due to the display overlay. If you are looking for displays for board or panel indication, telecom switches and central station equipment, control panels with high-brightness lighting, these compact

solutions might be the answer.

Developing displays for embedded systems

There are various standard low-cost display development kits available to aid your designs. These kits not only have the required hardware like display module, cables, controllers and power supply but also the reference manuals and codes (if required) to program your modules. You can also have live online support for working with the kits. Some vendors provide an exclusive GUI web-portal development site license along with their kits to help design engineers with an advanced approach to web-based development.

Using the latest hot pluggable USB-based OLED development kits, you can drive OLED displays from the USB port of a PC. These are extremely easy to use yet powerful demonstration tools that do not require any extra cable or power supply to run, allowing displays to be up and running in minutes. A kit will provide you with a USB controller card, mini USB cable, an interchangeable OLED display card and a CD with software application and drivers. Also available are the display expansion boards and capes for various development board environments. Besides, some vendors provide customised solutions with kit modules to quickly interface the dis-